



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

01-05-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C,D	A,B,D	C,D	A,B	A,B,C,D	A,B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	D	D	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.16 to 0.19	0.11 to 0.13	0.40	0.50	2.40	6.05 to 6.06	4.40 to 4.60	0.27 to 0.29

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C	A,B,C,D	A,C	A,B	A,B,C,D	D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	D	A	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.00	3.00	10.00	138.00	4.00	1.00	348.16	40.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	A,B,C	A,C,D	A,B	A,B,D	A,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	B	C	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.00	13.00	7.00	1.00	8.00	3.00	8.00	4.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B,C,D)

If they collide, their vertical component of velocities should be same,

$$\text{i.e., } 100\sin\theta = 160\sin 30^\circ$$

$$\Rightarrow \sin\theta = 4/5$$

Their vertical components will always be same.

Horizontal components.

$$\text{And } 160\cos 30^\circ = 80\sqrt{3} \text{ ms}^{-1}$$

They are not same, hence their velocities will not be same at any time. So (b) is correct.

$$x = x_1 - x_2 = 160\cos 30^\circ t - 100\cos\theta t$$

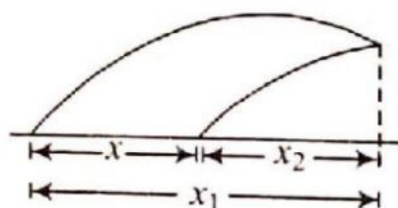
$$\text{Time of flight: } T = \frac{2 \times 160 \times \sin 30^\circ}{g}$$

$$= 16\text{s (same for both)}$$

Now $t < T_x \rightarrow$ to collide in air

$$\Rightarrow \frac{x}{80\sqrt{3} - 60} < 16 \Rightarrow x < 1280\sqrt{3} - 960$$

Since their times of flight are same, they will simultaneously reach their maximum height. So it is possible to collide at the highest point for certain values of x .



2. Ans (A,B,D)

According to Pascal's law, $P_D = P_B$

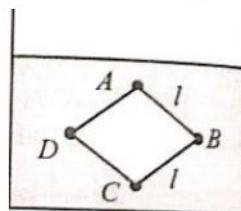
Because B and D lie on the same horizontal plane.

Let ℓ be the side of the square. Then the vertical distance between A and C is $\frac{\ell}{\sqrt{2}}$

$$P_B = P_V = P_C - \rho g \frac{L}{\sqrt{2}} \dots\dots (i)$$

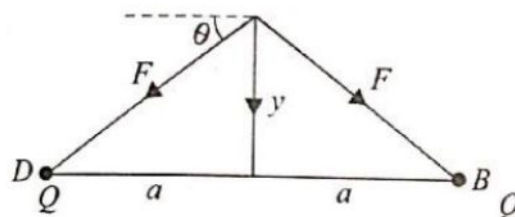
$$\text{Similarly, } P_B = P_D - P_A + \rho g \frac{L}{\sqrt{2}} \dots\dots\dots (ii)$$

$$P_B = P_D = \frac{P_A + P_C}{2}$$



3. Ans (C,D)

The particle will oscillate or perform SHM if equilibrium is a stable one. For negative charge, equilibrium is stable, if the particles are displaced along CD and unstable for displacement along AB.



For $y \ll a$, in displaced position, net resultant force toward equilibrium position is

$$F_{\text{res}} = 2F \sin\theta = \frac{2 \times Q}{4\pi\epsilon_0(a^2 + y^2)} \times \frac{y}{(a^2 + y^2)^{\frac{1}{2}}}$$

$$= \frac{2Qy}{4\pi\epsilon_0 a^3} [a \gg y]$$

$F_{\text{res}} \propto y$, so SHM for $y \ll a$

For $y = a$, there would be restoring force, but F is not proportional to y . So there is oscillating motion but not SHM.

4. Ans (A,B)

Let d = distance of the target T from the point of projection. P will strike T if d is an integral multiple of the pitch.

$$\text{Pitch} = \left(\frac{2\pi m}{QB} \right) v \cos \theta = k \left(\frac{v}{B} \right)$$

Where k = constant

$$\text{Initially, pitch } d = k \left(\frac{v_0}{B_0} \right)$$

For the given options if pitch $d' = \frac{d}{2}, \frac{d}{3}, \frac{d}{4}, \dots$

etc, then change will hit the target

5. Ans (A,B,C,D)

Δx at $O = d$ [path difference is maximum at O]

So, if $d = \frac{7\lambda}{2}$, O will be a minima and hence intensity is minimum.

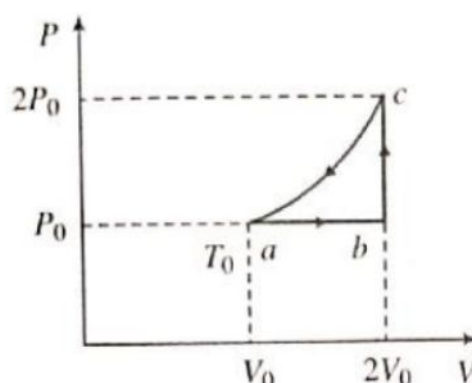
If $d = 4.8\lambda$, then total 10 minima can be observed on the screen, 5 above O and 5 below

O , which correspond to

$$\Delta x = \pm \frac{\lambda}{2}, \pm \frac{3\lambda}{2}, \pm \frac{5\lambda}{2}, \pm \frac{7\lambda}{2}, \pm \frac{9\lambda}{2}$$

6. Ans (A,B,C,D)

One mole of an ideal monoatomic gas (initial temperature T_0) is made to go through the cycle $abca$ shown in figure. U denotes the internal energy.



For the process ab , $\frac{P_0 V_0}{T_0} = \frac{2P_0 V_0}{T_b}$

$$T_b = 2T_0$$

$$T_b > T_a \Rightarrow U_b > U_a$$

$$U_b - U_a = C_V \Delta T = \frac{3R}{2} (2T_0 - T_0) = \frac{3RT_0}{2}$$

For the process bc ,

$$\frac{P_0(2V_0)}{2T_0} = \frac{2P_0(2V_0)}{T_c} \Rightarrow T_c = 4T_0$$

$$T_c > T_0$$

$$U_c - U_b = \frac{3R}{2} (4T_0 - T_0) = \frac{9RT_0}{2}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (A)**

Let the acceleration of various blocks are as

shown in Figure. Pulley P_2 will have

downward acceleration a .

$$\text{Now } a = \frac{a_1 + a_2}{2} \Rightarrow a_2 = 2a - a_1 > 0$$

So acceleration of 2 is upwards

Hence, (i) $\rightarrow$ (b, c)

$$\text{And } a = \frac{-a_1 + a_4}{2} \Rightarrow a_4 = 2a + a_1 > 0$$

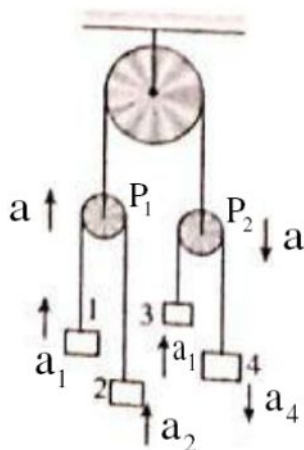
So, acceleration of 4 is downwards.

Hence, (ii) $\rightarrow$ (a, d)

Acceleration of 2 w.r.t 4

$$a_{\frac{2}{4}} = a_2 - (-a_4) = 4a > 0$$

This is upwards, Hence, (iv) $\rightarrow$ (c)



9. **Ans (D)**

$$(P) n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3$$

$$= 10 : 80 : 10 = 0.5 : 4 : 0.5$$

$$\Rightarrow \frac{5\lambda}{2} = 1 \Rightarrow \lambda = 0.4\text{m}$$

$$V = \sqrt{\frac{Y}{\rho}} = \frac{5000}{0.4} = 12500 = 12.5 \text{ kHz}$$

$$(Q) n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3$$

$$= 30 : 40 : 30 = 1.5 : 2 : 1.5$$

$$\Rightarrow \frac{5\lambda}{2} = 1 \Rightarrow \lambda = 0.4\text{m}$$

$$V = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{1.6 \times 10^{11}}{2500}} = 8000\text{m/s}$$

$$f = \frac{v}{\lambda} = \frac{8000}{0.4} = 20000 = 20 \text{ kHz}$$

$$(R) n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3$$

$$= 10 : 60 : 30 = 0.5 : 3 : 1.5$$

$$\Rightarrow \frac{5\lambda}{2} = 1 \Rightarrow \lambda = 0.4\text{m}$$

$$V = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{2 \times 10^{11}}{8000}} = 5000\text{m/s}$$

$$f = \frac{v}{\lambda} = \frac{5000}{0.4} = 12500 = 12.5 \text{ kHz}$$

$$(S) n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3$$

$$= 15 : 60 : 25 = 1.5 : 6 : 2.5$$

$$\Rightarrow \frac{10\lambda}{2} = 1 \Rightarrow \ell = 0.2\text{m}$$

$$V = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{1.6 \times 10^{11}}{2500}} = 8000 \text{ m/s}$$

$$f = \frac{v}{\lambda} = \frac{8000}{0.2} = 40000 = 40 \text{ kHz}$$

10. Ans (D)

Maximum kinetic energy of ejected electron is given by Einstein's photoelectric equation $K_{\max} = hf - \left(\frac{hf}{3}\right) = \frac{2hf}{3}$. As no potential difference is applied across target and collector and vacuum is there in the tube, so this maximum KE remains same at all locations. b, c and d. Kinetic energy of ejected photoelectrons can be anything from 0 to $K_{\max}$ (as found for a). It remains the same at all locations (reasoning is same as for above).

PART-1 : PHYSICS

SECTION-II

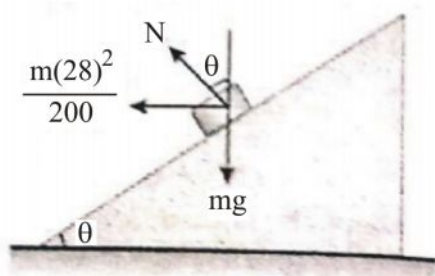
1. Ans (0.16 to 0.19)

When there is not tendency to slip, no frictional force acts. Hence

$$N \sin \theta = \frac{m(28)^2}{200} \dots\dots 1$$

$$\text{and } N \cos \theta = mg \dots\dots 2$$

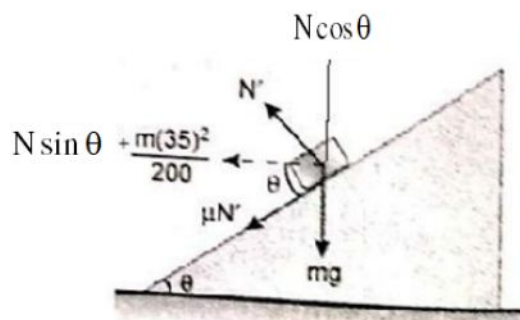
Dividing (1) by (2), we get $\tan \theta = 0.4 \dots\dots 3$



At speed 35 m/s, car has a tendency to slip up the track. Hence frictional force acts down the plane.

$$\text{Therefore, } N' \sin \theta + \mu N' \cos \theta = \frac{m(35)^2}{200} \dots\dots 4$$

$$\text{and } N' \cos \theta - \mu N' \sin \theta = mg \dots\dots 5$$



Solving (3), (4) and (5), we get $\mu = 0.18$

2. Ans (0.11 to 0.13)

$$\text{Initial kinetic energy } \frac{1}{2} \times 2 \times 10^2 = 100 \text{ J}$$

$$\text{Final KE} = 50 \text{ J}$$

$$\therefore \frac{1}{2}mv^2 = 50 \text{ J}$$

When the spring is being compressed, the equation of motion of spring mass system can be written as

$$x = A \sin \omega t \text{ where } \omega = \sqrt{\frac{k}{m}}$$

$$\therefore \frac{dx}{dt} = A \omega \cos \omega t$$

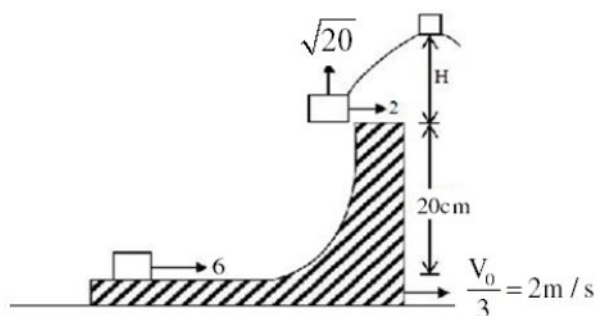
$$\Rightarrow v = v_0 \cos \omega t$$

$$\Rightarrow 5\sqrt{2} = 10 \cos \omega t$$

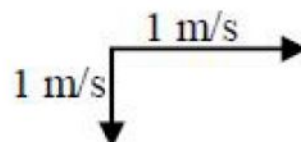
$$\Rightarrow \omega t = \frac{\pi}{4}$$

$$\Rightarrow t = \frac{\pi}{4\omega} = \frac{\pi}{4} \sqrt{\frac{m}{k}} = \frac{\pi}{4} \sqrt{\frac{2}{80}} = 0.124 \text{ sec}$$

3. Ans (0.40)



At highest point combined mass velocity is



$$H = \frac{20}{2g} = 1 \text{ m}$$

$$1.2 = 1.T + \frac{1}{2}gT^2$$

$$T = 0.4 \text{ sec}$$

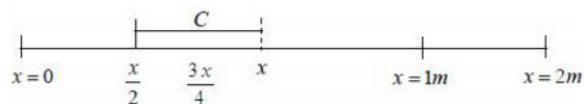
$$u_{\text{rel}} = 1 \text{ m/s}$$

$$\Rightarrow \ell = u_{\text{rel}} T = 0.4 = 40 \text{ cm}$$

4. Ans (0.50)

The co-ordinates of centre of mass (point C) of

the moving part is $\frac{3x}{4}$



$$\therefore X_c = \frac{3x}{4}$$

$$\therefore v_c = \frac{d}{dt}(X_c) = \frac{3}{4} \frac{dx}{dt} = \frac{3}{4} \text{ m/s} \left(\text{as } \frac{dx}{dt} = 1 \text{ m/s} \right)$$

Linear momentum of moving point is $P = mv$

Where $v = 1 \text{ m/s}$ and m is increasing uniformly

with time.

The net force on the moving part is thus,

$$F = \frac{dP}{dt} = \frac{dm}{dt}v + \frac{dv}{dt}m$$

$$\text{or } F = \left(\frac{dm}{dt} \right) (1) + 0$$

Here $\frac{dm}{dt}$ can be found with the help of following argument.

The moving end of the carpet starts from the

origin and the whole carpet will be moving

when it reaches $x = 2 \text{ m}$. This will happen after

$$2 \text{ seconds. Hence } \frac{dm}{dt} = \frac{1}{2} \text{ kg/s}$$

$$\therefore F_{\min} (\text{neglecting all dissipative forces}) = \frac{1}{2} \text{ N}$$

5. Ans (2.40)

Bulk modulus is given by $B = -\frac{dP}{(dV/V)}$ or

$$\frac{dV}{V} = -\frac{dP}{B}$$

$$\text{or } P \cdot \frac{dV}{V} = -\frac{P dP}{B}$$

$\frac{P dV}{V}$ is the work done per unit volume. The

negative sign implies that a decrease in pressure

gives rise to increase in volume and vice-versa.

Hence volume density of elastic potential

$$\text{energy } u = \int_0^{hpg} \frac{P \cdot dP}{B} = \frac{1}{2} \frac{(hpg)^2}{B}$$

Substitution the value, we obtain

$$u = \frac{1}{2} \frac{(10^3)(10^3)^2 \cdot (9.8)^2}{2 \times 10^9}$$

$$\Rightarrow u = 2.4 \times 10^4 \text{ J/m}^3$$

6. Ans (6.05 to 6.06)

Applying Snell's law on face AB,

$$1 \sin 45^\circ = \sqrt{2} \sin r$$

$$\therefore \sin r = \frac{1}{2}$$

$$\text{or } r = 30^\circ$$

i.e., ray becomes parallel to AD inside the block.

Now applying, $\frac{\mu_0}{v} = \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$ on face

$$CD = \frac{1.514}{OE} - \frac{\sqrt{2}}{\infty} = \frac{1.514 - 1.414}{0.4}$$

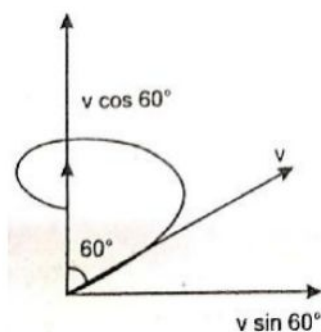
Solving this equation, we get $OE = 6.06 \text{ m}$

7. **Ans (4.40 to 4.60)**

Pitch of the helix is given by

$$y = (v \cos \theta) \cdot T = v(\cos 60^\circ)T = \frac{vT}{2}$$

Where time period of circular motion $T = \frac{2\pi m}{eB}$



$$\text{Pitch } y = \left(\frac{2\pi m}{Be} \right) v \cos 60^\circ = \left(\frac{\pi m v}{Be} \right)$$

$$\therefore KE = \frac{1}{2}mv^2 = \frac{1}{2}m \left(\frac{Be y}{\pi m} \right)^2$$

Using Einstein's equation

$$\frac{1}{2}mv^2 = hv - \phi$$

$$\phi = hv - \frac{1}{2}m \left(\frac{Be y}{\pi m} \right)^2$$

$$= \left[4.9 - \frac{1}{2} \left(\frac{m}{e} \right) \left(\frac{Be y}{\pi m} \right)^2 \right]$$

$$\text{electron - volts} = 4.9 - 0.4 = 4.5 \text{ eV}$$

8. **Ans (0.27 to 0.29)**

The product nucleus ^{198}Hg is in excited state and possesses extra 1.088 MeV energy.

If ^{198}Hg would had been in ground state, the kinetic energy available to electron and antineutrino must have

$$Q = (m_{\text{Au}} - m_{\text{Hg}}) 931 \text{ MeV}$$

$$= (197.968233 - 197.966760) 931 \text{ MeV}$$

$$= 1.3714 \text{ MeV}$$

Since ^{198}Hg is in excited state, actual kinetic energy available to electron and antineutrino is

$$K = (1.3714 - 1.088) \text{ MeV}$$

$$= 0.283 \text{ MeV}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (A,B,C)**

Metals of 2nd & 3rd transition series form low spin complexes with weak or strongfield ligand and form square planar complexes with complex number 4.

Ni(II) with strong ligand like CN^- form square planar complexes.

Hg(II) form tetrahedral complexes since it has d^{10} config.

2. **Ans (A,B,C,D)**

Froth floatation method is a physical method depending on adsorption of potassium ethyl xanthate which prevents the wettability of or particle. It is used only for the concentration of sulphide ores where ore particles come into frothwhile impurities sink down.

3. **Ans (A,C)**

Both $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ are not possible on vinyl halides and aryl halides.

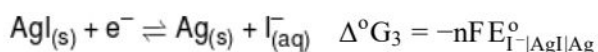
4. **Ans (A,B)**

Value based.

5. **Ans (A,B,C,D)**

Variation of composition changes internal energy of the system.

6. **Ans (D)**



$$\Delta^\circ G_3 = \Delta^\circ G_1 + \Delta^\circ G_2$$

Substituting, we get

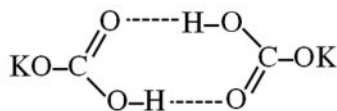
$$\log K_{\text{sp}} = \frac{-0.952}{0.059} = -16.13$$

PART-2 : CHEMISTRY

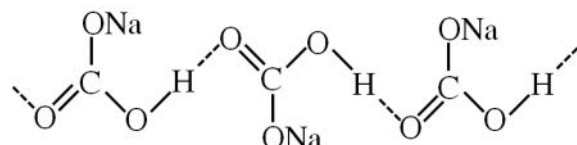
SECTION-I (ii)

7. **Ans (A)**

KHCO_3 exist as dimer.



NaHCO_3 exist in polymeric form.



LiHCO_3 exist only solution and exist as Li^+ and

HCO_3^- similar to $\text{Mg}(\text{HCO}_3)_2$

(diagonal relationship).

8. **Ans (D)**

(A) $\text{H}_2\text{P}_2\text{O}_6^{2-}$ have 2 ionisable hydrogen. The phosphorous atoms are involved in sp^3 hybridization. It is non-planar species.

(B) Similarly $\text{H}_2\text{P}_2\text{O}_7^{2-}$ also have 2 ionisable hydrogens, sp^3 hybrid p atoms and non-planar structure.

(C) $\text{H}_2\text{S}_2\text{O}_5$ have 2 ionisable hydrogens, sp^3 hybrid sulphur atoms with non-planar structure.

(D) HSO_5^- ion has only one ionizable hydrogen in which S-atom is involved in sp^3 hybridization with non-planar structure. It can act as oxidizing agent.

9. **Ans (A)**

Direct reactions.

10. **Ans (B)**

All are fcc lattices. Symmetry is followed.

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (1.00)**

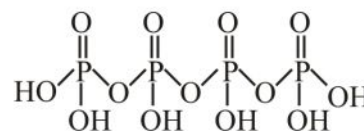
NaOH , KOH , $\text{Ba}(\text{OH})_2$, $\text{Sr}(\text{OH})_2$ and TlOH are strong bases. $\text{Be}(\text{OH})_2$, $\text{Al}(\text{OH})_3$, $\text{Sn}(\text{OH})_2$ and $\text{Zn}(\text{OH})_2$ are amphoteric and these react with each other forming salts that are soluble.

$\text{Mg}(\text{OH})_2$ is insoluble in bases. So only $\text{Mg}(\text{OH})_2$ remain in the residue. One base is excess and that also dissolve in water.

2. **Ans (3.00)**

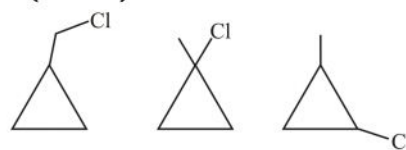
The ratio of the two different types of OH groups should be determined. If one phosphorous atom is attached by two OH groups, their K_{a1} and K_{a2} values differ.

The first OH group is a strongly acidic while the second one is weak. If only one OH group is present on one phosphorous atom. It also behave strongly acidic. So phosphorous atoms which contain two OH groups will ionize in two steps. The strongly acidic groups are titrated by the first 30.4. The two terminal OH groups are titrated by the remaining $45.6 - 30.4 = 15.2$ CC. This indicates each OH requires 7.6 CC of the titrant. It can be now calculated that there are titrant. It can be now calculated that there are $\frac{30.4}{7.6} = 4$ strongly acidic groups per one polyphosphoric acid. A molecule with two terminal OH groups and four further OH groups is a tetrapolyphosphoric acid.



Thus the number of P - O - P bonds is 3.

3. **Ans (10.00)**

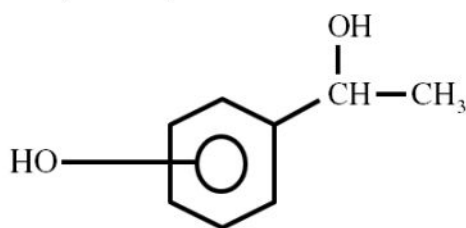


$$P \quad 1 \quad + \quad 1 \quad + \quad 4 = 6$$

$$Q \quad 1 \quad + \quad 1 \quad + \quad 2 = 4$$

$$P + Q = 10$$

4. Ans (138.00)



∴ minimum molecular weight = 138

5. Ans (4.00)

I, IV, VI, VII

6. Ans (1.00)



t=0 2 mol 0.5 mol 0

t=t 1.5 mol 0.25 mol 0.5 mol

$$\Delta P = \frac{0.25 \times RT}{V} = \frac{0.25}{12} \times \frac{300}{6.25} = 1 \text{ atm}$$

7. Ans (348.16)

$$\begin{aligned} \text{IE}_4(\text{Be}) &= \frac{Z^2}{n^2} \times 13.6 \times 1.6 \times 10^{-19} \text{ J} \\ &= 348.16 \times 10^{-19} \end{aligned}$$

8. Ans (40.00)

$$\text{Average rate in first 10 min.} = \frac{\Delta p}{\Delta t} = \frac{2}{10} \frac{\text{atm}}{\text{min}}$$

$$\text{Average rate in next 10 min.} = \frac{\Delta p}{\Delta t} = \frac{2}{10} \frac{\text{atm}}{\text{min}}$$

∴ Reaction is zero order

$$\therefore t = \frac{8}{K} = \frac{8}{1/5} = 40 \text{ mins}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B,C,D)

$$\int_1^{xy} f(t)dt = y \int_1^x f(t)dt + x \int_1^y f(t)dt, f(1) = 1$$

Differentiating partially w.r.t. 'y'

$$f(xy).x = \int_1^x f(t)dt + xf(y), \text{ , put } y = 1$$

$$\Rightarrow x f(x) = \int_1^x f(t)dt + x$$

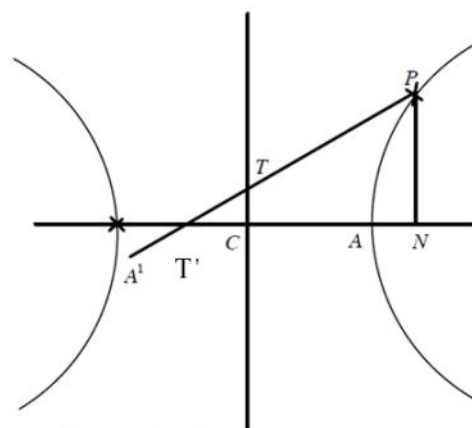
$$\Rightarrow f'(x) = \frac{1}{x}$$

$$\Rightarrow f(x) = \begin{cases} \ln(-x) + c, & x < 0 \\ \ln x + 1, & x > 0 \end{cases}$$

∴ A, B, C, D are true

2. Ans (A,B,C)

$$C=(0, 0) \quad N = (\text{asec } \theta, 0)$$



$$\text{Clearly } T = \left(0, -\frac{b}{\tan \theta}\right)$$

$$T' = \left(\frac{a}{\sec \theta}, 0\right)$$

$$CT.PN = b^2$$

$$CN.CT^1 = a^2$$

$$CN.NT^1 = AN.A^1N = a^2 \tan^2 \theta$$

∴ A,B,C are correct

3. Ans (A,C,D)

a, b, c are integers and $a^2 + b^2 + c^2 = 2 \Rightarrow$ one of them is 0 and the other can be ± 1 .

$\therefore$ No. of cases = 12

$$\text{For unique solution } |A| \neq 0 \Rightarrow \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \neq 0$$

$\therefore a = 0, b = 1, c = -1$ and there are 6 cases

$$\therefore \text{Probability} = \frac{6}{12} = \frac{1}{2}$$

But in all the remaining 6 cases it cannot have infinitely many solutions

For example for $a = 0, b = -1, c = 1, 1 \Delta_1 \neq 0$

$\therefore$ It has no solution

$\therefore$ B is not correct

$a = 0 \Rightarrow b^2 + c^2 = 2 \quad (1, -1) (-1, 1)$ it has no solution

$$a^2 + b^2 + c^2 - ab - bc - ca \geq 0$$

$$\Rightarrow ab + bc + ca \leq 2$$

$$(a + b + c) \geq 0 \Rightarrow ab + bc + ca \geq -1$$

But $a = b = c$ not possible

$\therefore 2$ is not there

$$ab + bc + ca \neq 0$$

$\therefore -1, 1$ are possible values

4. Ans (A,B)

$$\text{The plane after rotation is } \begin{vmatrix} x-3 & y-1 & z-2 \\ 2 & 4 & 4 \\ 1 & 4 & 4 \end{vmatrix} = 0$$

$$\Rightarrow (x-3)(12) - (y-1)(6) + (z-2)(0) = 0$$

$$\Rightarrow 2x - 6 - y + 1 = 0$$

$$\Rightarrow 2x - y - 5 = 0$$

Equation of plane before rotation

$$\Rightarrow \begin{vmatrix} x-3 & y-1 & z-2 \\ 1 & 2 & 2 \\ 2 & -1 & 0 \end{vmatrix} = 0$$

$$\Rightarrow 2x + 4y - 5z = 0$$

5. Ans (A,B,D)

From graph clearly it will have a solution when $a > 0$ and when $a < 0$ then

$$\Rightarrow x^2 - a = x \text{ for atleast one } x$$

$$\Rightarrow 1 + 4a \geq 0 \Rightarrow a \geq -\frac{1}{4}$$

$\therefore A, B, D$ is the key

6. Ans (A,C)

For $\Delta A_1 B_1 C_1$ the triangle ABC will be the pedal triangle

$\therefore$ Orthocentre of $\Delta A_1 B_1 C_1$ will be the incentre of ΔABC

$$A_1 = 90^\circ - \frac{A}{2} = \frac{\pi}{3} - \left(\frac{A}{2} - \frac{\pi}{6} \right)$$

$$A_2 = \frac{\pi}{3} - \left(-\frac{A}{2} + \frac{\pi}{12} \right)$$

$$\therefore \lim_{x \rightarrow \infty} A_n = \frac{\pi}{3}$$

$\therefore \Delta A_n B_n C_n$ becomes equilateral

$$B_1 C_1 = R \sin \left(90^\circ - \frac{A}{2} \right) = R \cos \frac{A}{2} = \frac{a}{2 \sin A} \cdot \cos \frac{A}{2} \\ = \frac{a}{2 \cdot 2 \sin \frac{A}{2} \cos \frac{A}{2}} \cos \frac{A}{2} = \frac{a}{4} \cos \frac{A}{2}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A)

$$\overline{SA} + \overline{SB} + \overline{SC} = \overline{OA} + \overline{OB} + \overline{OC} - 3\overline{OS} = 3\overline{OG} - 3\overline{OS} = 3(\overline{SG})$$

$A \rightarrow Q$

$$\overline{OA} + \overline{OB} + \overline{OC} = 3\overline{OG}$$

$$3G = 2O + O' \quad \begin{array}{c} | \quad | \quad | \\ O' \quad \quad G \quad O \end{array}$$

$$|\overline{OO'}| = 3|\overline{OG}|$$

$$\overline{O'A} + \overline{O'B} + \overline{O'C} = \overline{OA} + \overline{OB} + \overline{OC} - (\overline{OO'} + \overline{OO'} + \overline{OO'}) \\ = 3\overline{OG} - 3\overline{OO'} = \overline{OO'} - 3\overline{OO'} = 2\overline{OO'}$$

$\therefore$ Answer is 2.

$$D) |\overline{OO'} - \overline{OA} + \overline{OB} - \overline{OO'} + \overline{OC} - \overline{OO'}|$$

$$|\overline{OA} + \overline{OB} + \overline{OC} - 2\overline{OA} - \overline{OO'}| = |$$

$$3\overline{OG} - 2\overline{OA} - \overline{OO'} = |2\overline{OA}| = 2R = |\overline{AP}|$$

$\therefore$ Answer is 5.

8. Ans (B)

$$(A) \begin{vmatrix} \bar{a} & \bar{b} & \bar{c} \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 2 & -1 \end{vmatrix}^2$$

$$= (1(-1) - 1(-2) + 1(3))^2 = 16$$

A → r

$$B) \text{ Equation of OQR is } \begin{vmatrix} x & y & z \\ 1 & 3 & 4 \\ 2 & 1 & -2 \end{vmatrix} = 0$$

$$\Rightarrow x(-10) - y(-10) + z(-5) = 0$$

$$\Rightarrow 2x - 2y + z = 0$$

$$\therefore \text{Distance} = \frac{|16 + 4 - 11|}{3} = \frac{9}{3} = 3$$

C) $(\lambda, \lambda-1, \lambda), (2u-1, u, u)$ be A and B

$\therefore$ Dr's of AB are $(2u-\lambda-1, u-\lambda+1, u-\lambda)$ (2, 12)

$$\frac{2u-\lambda-1}{2} = \frac{u-\lambda+1}{1} = \frac{u-\lambda}{2}$$

$$\Rightarrow \lambda = 3$$

$$2u - 2\lambda + 2 = u - \lambda$$

$$u = \lambda - 2 = 1$$

$$A(3, 2, 3) \quad B(1, 1, 1)$$

$$\therefore AB = \sqrt{4+1+4} = 3$$

D) Let OABC be the tetrahedron

$$\text{Centre of } \overline{OAB} = \frac{\bar{a} + \bar{b}}{3} = G_1. \quad \text{Centre of}$$

$$\overline{OBC} = \frac{\bar{b} + \bar{c}}{3} = G_2$$

$$\therefore G_3 = \frac{\bar{c} + \bar{a}}{3}, G_4 = \frac{\bar{a} + \bar{b} + \bar{c}}{3}$$

$\therefore$ Volume of smaller tetrahedron

$$= \frac{1}{6} \left| \begin{vmatrix} \bar{c} & \bar{a} & \bar{b} \\ 3 & 3 & 3 \end{vmatrix} \right|$$

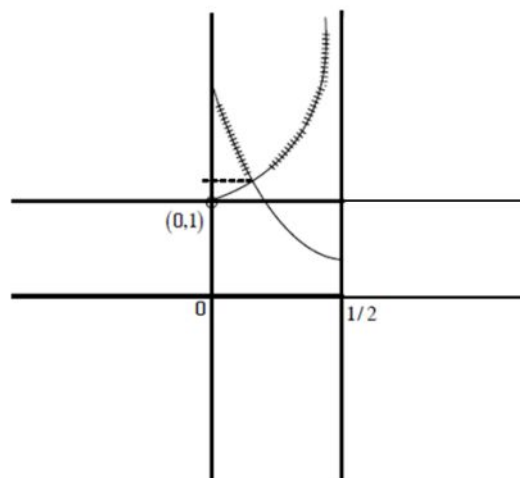
$$\text{Volume of tetrahedron} = \frac{1}{6} |[\bar{a} \quad \bar{b} \quad \bar{c}]|$$

$$\therefore \text{Ratio} = \frac{1}{27}$$

$$\therefore m + n = 28$$

9. Ans (C)

$$0 < x < \frac{1}{2} \Rightarrow 0 < \pi x < \frac{\pi}{2} \Rightarrow 0 < \sin \pi x < 1$$



They intersect at $x = \frac{1}{4}$

$$\therefore \text{Range} = (3 - \sqrt{2}, 2] - \{1\}$$

Numbers integers = 1

Clearly no. of extreme = 1

$$\text{At } x = \frac{1}{4}$$

$f(x)$ is an integer

10. Ans (C)

A) $\sqrt{\sin x}$ is increasing on $\left[0, \frac{\pi}{3}\right]$ so it is one-one

$$\text{and range} = \left[0, \sqrt{\frac{\sqrt{3}}{2}}\right]$$

$\therefore$ into

$$B) f(x_1) = f(x_2) \Rightarrow \frac{x_1 + 3}{x_1 - 1} = \frac{x_2 + 3}{x_2 - 1}$$

$$\Rightarrow -x_1 + 3x_2 - 3 = 3x_1 - 3 - x_2$$

$$\Rightarrow 3(x_1 - x_2) + (x_1 - x_2) = 0 \Rightarrow x_1 = x_2$$

$$y = \frac{x+3}{x-1} \Rightarrow xy - y = x + 3$$

$$\Rightarrow x = \frac{y+3}{y-1}$$

$$x > 1 \Rightarrow \frac{y+3}{y-1} - 1 > 0$$

$$\Rightarrow \frac{4}{y-1} > 0 \Rightarrow y > 1$$

$\therefore$ Range = $(1, \infty)$

$\therefore$ One-One, onto

$$C) \sin x \text{ is one-one } \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$\therefore$ It is many-one onto

$$f'(x) = \frac{x(x-4)}{(x-2)^2}$$

$\therefore$ It is $\uparrow$ on $(4, \infty)$ and $\downarrow$ in $(2, 4)$

$\therefore$ It is many-one

$$D) y = \frac{x^2}{x-2} \Rightarrow x^2 - xy + 2y = 0$$

$$\Rightarrow y^2 - 8y \geq 0$$

$$\Rightarrow \text{but } y > 0$$

$$\therefore y \in [8, \infty)$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1.00)

$$\arg(z^{1/3}) = \frac{1}{2} \arg(z^2 + \bar{z}z^{1/3})$$

$$\arg\left(\frac{z^{1/3}}{z^2 + \bar{z}z^{1/3}}\right) = 0$$

$$\Rightarrow \frac{z^{2/3}}{z^2 + \bar{z}z^{1/3}} = \frac{\bar{z}^{2/3}}{\bar{z}^2 + z\bar{z}^{1/3}}$$

$$\Rightarrow z^{2/3} \cdot \bar{z}^2 + z\bar{z}^{2/3} z^{-1/3} = \bar{z}^2 \bar{z}^{2/3} + \bar{z}^{5/3} z^{1/3}$$

$$\Rightarrow z^{2/3} \cdot \bar{z}^{2/3} (\bar{z}^{4/3} - z^{4/3}) + z^{1/3} \bar{z}^{1/3} (z^{4/3} - \bar{z}^{4/3}) = 0$$

$$\Rightarrow (\bar{z}^{4/3} - z^{4/3})(|z|^{2/3} - |z|^{4/3}) = 0$$

$$\Rightarrow z = \bar{z} \Rightarrow z \text{ is real not possible}$$

$$\therefore |z|^{2/3}(|z|^{2/3} - 1) = 0 \Rightarrow |z| = 1$$

2. Ans (13.00)

$$\text{Equation of circle is } x^2 + y^2 = a^2 e^2,$$

$$\text{Let } P = (ae \cos \theta, ae \sin \theta)$$

$$\text{Area of triangle} = 30 \Rightarrow a^2 e^2 \sin \theta = 30 \quad \dots (1)$$

$$\text{But } P \text{ lies on the ellipse} \Rightarrow \sin \theta = \frac{1}{e^2} - 1$$

$$\therefore (1) \Rightarrow \frac{289}{4} e^2 \frac{(1 - e^2)}{e^2} = 30$$

$$\Rightarrow e^2 = \frac{169}{289} \Rightarrow e = \frac{13}{17}$$

$$\therefore S_1 S_2 = 2 \times \frac{17}{2} \times \frac{13}{17} = 13$$

3. Ans (7.00)

Given

$$\det = \begin{vmatrix} -1 & b & a+2b+2c \\ -1 & c & b+2a+2c \\ -1 & a & c+2a+2b \end{vmatrix} \quad c_1 \rightarrow c_1 + c_2$$

$$= \begin{vmatrix} -1 & b & a+2b+2c \\ 0 & c-b & a-b \\ 0 & a-c & b-c \end{vmatrix}$$

$$= a^2 + b^2 + c^2 - ab - bc - ca$$

$$= (-1)^2 - 3(-2) = 7$$

4. Ans (1.00)

$$a_2 = 2(1 + a_1)$$

$$\Rightarrow 1 + a_1 = \frac{a_2}{2}, 1 + a_2 = \frac{a_3}{3} \dots \dots 1 + a_{n-1} = \frac{a_n}{n}$$

Given limit =

$$\lim_{n \rightarrow \infty} \left(\frac{1+a_1}{a_1} \right) \left(\frac{1+a_2}{a_2} \right) \dots \dots \left(\frac{1+a_n}{a_n} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{2} \cdot \frac{1}{3} \dots \dots \frac{1}{n} (1+a_n) = \lim_{n \rightarrow \infty} \frac{(1+a_n)}{n!}$$

$$\text{But } \frac{a_n}{n!} = \frac{1}{(n-1)!} + \frac{a_{n-1}}{(n-1)!}$$

$$\Rightarrow \frac{a_n}{n!} - \frac{a_{n-1}}{(n-1)!} = \frac{1}{(n-1)!}$$

$$\text{Put } n = 2, 3, 4 \dots \dots \frac{a_2}{2!} - \frac{a_1}{1!} = \frac{1}{1!}$$

$$\frac{a_3}{3!} - \frac{a_2}{2!} = \frac{1}{2!}$$

$$\frac{a_n}{n!} - \frac{a_{n-1}}{(n-1)!} = \frac{1}{(n-1)!}$$

$$\frac{a_n}{n!} - \frac{a_1}{1!} = \frac{1}{1!} + \frac{1}{2!} + \dots \dots + \frac{1}{(n-1)!}$$

$$\therefore \frac{a_n}{n!} = 1 + \frac{1}{1!} + \frac{1}{2!} + \dots \dots + \frac{1}{(n-1)!}$$

$$\therefore \lim_{n \rightarrow \infty} \left(\frac{1+a_n}{n!} \right) = \lim_{n \rightarrow \infty} \frac{1}{n!} + \lim_{n \rightarrow \infty} \frac{a_n}{n!} = e$$

$$\therefore \text{Required value} = e \times \frac{1}{e} = 1$$

5. Ans (8.00)

$$f(x) + f(y) \leq 0$$

$$\Rightarrow x^2 + 6x + 1 + y^2 + 6y + 1 \leq 0$$

$$\Rightarrow (x+3)^2 + (y+3)^2 \leq 16$$

$$f(x) - f(y)$$

$$\Rightarrow x^2 + 6x + 1 - y^2 - 6y - 1 \leq 0$$

$$\Rightarrow (x^2 - y^2) + 6(x - y) \leq 0$$

$$\Rightarrow (x - y)(x + y + 6) \leq 0$$

Required area is the area of semicircle = 8π

$$\therefore k = 8$$

6. Ans (3.00)

$$\cos^2 \frac{\pi}{8} = \frac{\sqrt{2}+1}{2\sqrt{2}}, \text{ the other root is } \frac{\sqrt{2}-1}{2\sqrt{2}}$$

$$\therefore \text{Sum} = 1 = -p \Rightarrow p = -1$$

$$\therefore [\cos 1 + \sin 1] = \left[\sqrt{2} \sin \left(\frac{\pi}{4} + 1 \right) \right] = 1$$

$$q^2 = \frac{1}{8} \Rightarrow 1 - 1 + 2 \left| \log_2 \frac{1}{2\sqrt{2}} \right| = 3$$

7. Ans (8.00)

$$\text{Let } \bar{a} = (\cos x \cos y, \sin y, \sin x \cos y), \bar{b} = (1, 2, 2)$$

$$\bar{a} \cdot \bar{b} = \cos x \cos y + 2 \sin y + 2 \sin x \cos y$$

$$(\bar{a} \cdot \bar{b})^2 \leq |\bar{a}|^2 |\bar{b}|^2$$

$$\Rightarrow (\cos x \cos y + 2 \sin y + 2 \sin x \cos y)^2 \leq 1.9$$

$$\Rightarrow \cos x \cos y + 2 \sin y + 2 \sin x \cos y \leq 3$$

$$\therefore \frac{\cos x \cos y}{1} = \frac{\sin y}{2} = \frac{\sin x \cos y}{2}$$

$$\tan y = 2 \cos x, \tan y = \sin x$$

$$\tan x = 2, \tan y = \frac{2}{\sqrt{5}}$$

$$\therefore \tan^2 x + 5 \tan^2 y = 4 + 4 = 8$$

8. Ans (4.00)

$$\lim_{n \rightarrow \infty} \left(\frac{3!}{2n!n!} \cdot \frac{n!n!}{(2n)!} \right)^{\frac{1}{n}} = \frac{a}{b}$$

$$y = \lim_{n \rightarrow \infty} \left(\frac{(2n+1)(2n+2) \dots (2n+n)}{(n+1)(n+2) \dots (n+n)} \right)^{\frac{1}{n}} = \frac{a}{b}$$

$$\lim_{n \rightarrow \infty} \left(\frac{\left(2 + \frac{1}{n}\right) \left(2 + \frac{2}{n}\right) \dots \left(2 + \frac{n}{n}\right)}{\left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \dots \left(1 + \frac{n}{n}\right)} \right)^{\frac{1}{n}} = \frac{a}{b}$$

$$\log y = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left[\log \left(2 + \frac{r}{n} \right) - \log \left(1 + \frac{r}{n} \right) \right] = \frac{a}{b}$$

$$= \int_0^1 \log \left(\frac{2+x}{1+x} \right) dx = \int_2^3 \log t dt - \int_1^2 \log z dz$$

$$= (t(\log t - 1))_2^3 - (z(\log z - 1))_1^2$$

$$= 3(\log 3 - 1) - 2(\log 2 - 1) - 2(\log 2 - 1) + 1(0 - 1)$$

$$= 3 \log 3 - 4 \log 2 = \log \frac{27}{16}$$

$$\therefore \frac{a}{b} = \frac{27}{16}$$

$$\frac{a+b+1}{a-b} = \frac{27+16+1}{27-16} = \frac{44}{11} = 4$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

01-05-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	C	B	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B,C	A,B,C	A,D	A,B,C	A,B,C	A,B,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	2	8	0	8	2	9	2	2

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	D	C	B	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C,D	A,B	A,B,C,D	B,C,D	A,B,D	A,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	3	4	4	8	7	4	6	8

PART-3 : MATHEMATICS

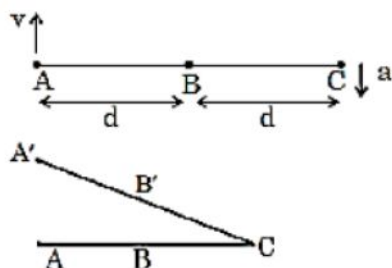
SECTION-I (i)	Q.	1	2	3	4				
	A.	D	B	D	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C	A,B,C	B,C	B,D	B,C,D	A,C,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	3	4	4	3	0	1	8	7

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. **Ans (A)**



Let us observe the motion of A and B relative to C.

$$\frac{AA'}{2d} = \frac{BB'}{d}$$

$$BB' = \frac{AA'}{2} = \frac{vt + \frac{1}{2}at^2}{2}$$

$$BB' = \frac{Vt}{2} + \frac{1}{4}at^2$$

$\frac{V}{2}$ is initially speed of B w.r.t C as well as ground.

$\frac{a}{2}$ is acceleration of B w.r.t C

$$\vec{a}_{B/g} = \vec{a}_{B/C} + \vec{a}_{C/g}$$

$$= \frac{a}{2} - a = -\frac{1}{2}a$$

4. **Ans (B)**

The wavelength at which the intensity of radiation is maximum depends only on the temperature. As this wavelength is the same for both, their temperatures must be equal. The difference in intensities can then be due only to a difference in their emissivities.

PART-1 : PHYSICS

SECTION-I (ii)

5. **Ans (B,C)**

Negative zero error = -1.25

$$\text{Thickness} = 18 + 0.34 - (-1.25) = 19.59 \text{ mm}$$

6. **Ans (A,B,C)**

$$\text{Speed} = v = \sqrt{\frac{T}{\mu}} = 50 \text{ m/s}$$

$$\text{Power} = \frac{1}{2} \mu \omega^2 a^2 v = 0.126 \text{ W}$$

$$E_{\text{avg}} = P_{\text{avg}} \cdot t = P_{\text{avg}} \left(\frac{1}{9} \right) = 2.5 \text{ mJ}$$

7. **Ans (A,D)**

$$\begin{aligned} U &= \int_R^{2R} \frac{1}{2} \epsilon_0 E^2 dV = \frac{1}{2} \epsilon_0 \frac{K^2 Q^2}{r^4} 4\pi r^2 dr \\ &= \frac{KQ^2}{2} \int_R^{2R} r^{-2} dr = \frac{Q^2}{8\pi\epsilon_0} \left[\frac{1}{-2R} + \frac{1}{R} \right] \\ &= \frac{Q^2}{16\pi\epsilon_0 R} \end{aligned}$$

Now no electric field inside sphere.

So $U = 0$ (inside)

$$U_{\text{self}} = \frac{KQ^2}{2R} = \text{heat lost}$$

8. **Ans (A,B,C)**

Potential difference is defined only for electrostatic field.

9. **Ans (A,B,C)**

$$\begin{aligned} \frac{n_p}{n_p \left(\frac{d}{n_p} + \frac{t}{1} + \frac{x}{n_g} \right)} + \frac{1}{\infty} &= \frac{(1 - n_p)}{-R} \\ \Rightarrow \frac{d}{n_p} + t &= \frac{R}{n_p - 1} - \frac{x}{n_g} \end{aligned}$$

10. Ans (A,B,C)

$$Q = \frac{h^2}{24m\lambda^2} = K_B + K_C$$

Where k is the kinetic energy

From conservation of linear momentum

$$|\vec{P}_C| = |\vec{P}_A| = P$$

$$\therefore K = \frac{P^2}{2m}$$

$$\therefore Q = \frac{P^2}{2 \times 12m} + \frac{P^2}{2 \times 4m}$$

$$\frac{h^2}{24m\lambda^2} = \frac{P^2}{2} \left[\frac{1}{12m} + \frac{1}{4m} \right]$$

$$\frac{h^2}{12m\lambda^2} = P^2 \left(\frac{1+3}{12m} \right) = \frac{P^2}{3m}$$

$$\Rightarrow P^2 = \frac{h^2 \times 3m}{12m\lambda^2} \Rightarrow P = \frac{h}{2\lambda}$$

$$\Rightarrow \frac{h}{P} = 2\lambda = \text{de-Broglie wavelength}$$

$$\therefore |\vec{P}_B| = |\vec{P}_C|$$

$$\text{Mass defect, } \Delta m = \frac{Q}{C^2} = \frac{h^2}{24mc^2\lambda^2}$$

$$\therefore m_A = 16m + \frac{h^2}{24mc^2\lambda^2}$$

PART-1 : PHYSICS

SECTION-III

1. Ans (2)

$$k = \frac{2\pi}{\lambda} = 62.8\lambda = 0.1m\ell = 40\frac{\lambda}{2} = 2m$$

2. Ans (8)

$$Kx \cos 37^\circ = T \cos 53^\circ$$

$$Kx \cos 53^\circ + T \cos 37^\circ = mg$$

$$\Rightarrow \frac{T (\cos^2 53^\circ + \cos^2 37^\circ)}{\cos 37^\circ} = mg$$

$$\Rightarrow T = \frac{4mg}{5}$$

$$K\vec{x} + \vec{T} + m\vec{g} = 0$$

$K\vec{x}$ does not change suddenly

$$\Rightarrow K\vec{x} + m\vec{g} = m\vec{a} = -\vec{T}$$

$$ma = \frac{4mg}{5}$$

$$\Rightarrow a = 8m/s^2$$

3. Ans (0)

When slider is at A, the applied potential difference

across the photoelectric cell is a retarding potential

difference of 10 V. Energy of incident photon

$$E = \frac{hc}{\lambda} = \frac{1240\text{eV nm}}{350\text{nm}} = 3.5\text{eV}$$

Maximum KE of emitted photoelectrons is

$$KE_{\max} = \frac{hc}{\lambda} - \phi = 3.5 - 2.2 = 1.3\text{eV}$$

stopping potential difference

$$V_s = 1.3\text{V}$$

(a) The applied potential difference is

$$V (= 10\text{V}) > V_s (= 1.3\text{V})$$

Hence there is no photocurrent and reading of

ammeter is zero.

4. Ans (8)

By the conservation of energy

$$\frac{1}{2}mV_0^2 = mgh + \frac{1}{2}mv^2 \dots \dots \dots (1)$$

By the conservation of angular momentum

$$\frac{mV_0r}{2} = mVR \dots \dots \dots (2)$$

From (1) & (2)

$$V_0 = \sqrt{\frac{8gh}{3}}$$

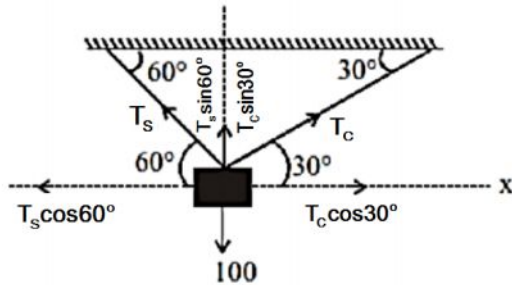
$$k = 8$$

5. Ans (2)

Let T_S = tension in steel wire

T_C = Tension in copper wire

In x direction



$$T_C \cos 30^\circ = T_S \cos 60^\circ$$

$$T_C \times \frac{\sqrt{3}}{2} = T_S \times \frac{1}{2}$$

$$\sqrt{3}T_C = T_S \dots\dots\dots (i)$$

In y direction

$$T_C \sin 30^\circ + T_S \sin 60^\circ = 100$$

$$\frac{T_C}{2} + \frac{T_S \sqrt{3}}{2} = 100 \dots\dots\dots (ii)$$

Solving equation (i) & (ii)

$$T_C = 50N$$

$$T_S = 50\sqrt{3}N$$

$$\text{We know } \Delta L = \frac{FL}{AY}$$

$$= \frac{\Delta L_C}{\Delta L_S} = \frac{T_C L_C}{A_C Y_C} \times \frac{A_S Y_S}{T_S L_S}$$

On solving above equation

$$\frac{\Delta L_C}{\Delta L_S} = 2$$

6. Ans (9)

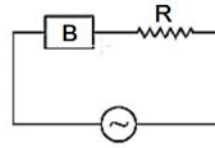
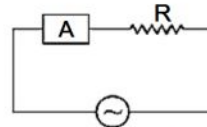
Since water resistance ($\vec{F} = -k\vec{v}_{pr}$) is proportional to the velocity of plank relative to the water, impulse of water resistance equals to product of the proportionality constant k and displacement of the plank relative to the water. And this impulse stops the block- plank system at the other bank, therefore equals to the initial momentum of the block.

$$mu_{\min} = k \int v dt$$

$$\Rightarrow mu_{\min} = k(b - l)$$

$$u_{\min} = \frac{k(b - l)}{m} = 9.0m/s$$

7. Ans (2)



$$\phi_1 = -30^\circ$$

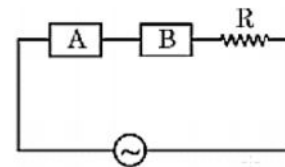
$$\phi_2 = +60^\circ$$

$$\tan \phi = \frac{X_L}{R}$$

$$\tan \phi_2 = \frac{X_C}{R}$$

$$X_L = \frac{R}{\sqrt{3}}$$

$$X_C = R\sqrt{3}$$



$$\tan \theta = \frac{(X_C - X_L)}{R} = \frac{\frac{R\sqrt{3}-R}{\sqrt{3}}}{R} = \frac{2}{\sqrt{3}}$$

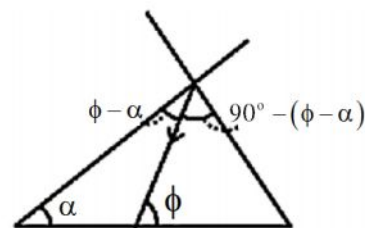
8. Ans (2)

$$a_1 \cos[90^\circ - (\phi - \alpha)] = a_2 \sin \alpha$$

$$a_1 \sin(\phi - \alpha) = a_2 \sin \alpha \dots\dots\dots (i)$$

Conservation has no acceleration x direction

$$\Rightarrow Ma_2 = ma_1 \sin \phi \dots\dots\dots (ii)$$



From (i) and (ii)

$$\Rightarrow \frac{m}{M} \sin \phi = \frac{\sin(\phi - \alpha)}{\sin \alpha}$$

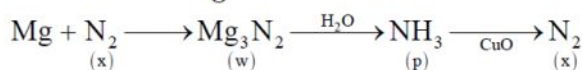
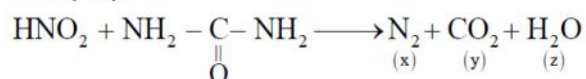
$$\Rightarrow \frac{m}{M} = \frac{\tan \phi - \tan \alpha}{\tan \alpha}$$

$$\frac{m}{M} = \frac{\tan 60 - \tan 30}{\tan 30} = 2$$

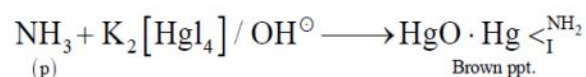
PART-2 : CHEMISTRY

SECTION-I (i)

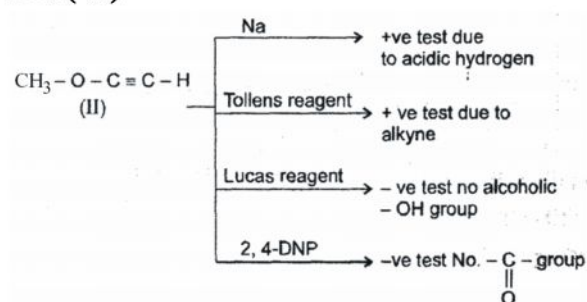
1. **Ans (D)**



Milky solution



2. **Ans (C)**



3. **Ans (B)**

From the equation,

$$[\text{Cu(CN)}_4^{3-}] = 0.1\text{M and } [\text{CN}^{-1}] = 0.2\text{M}$$

$$\therefore [\text{Cu}^+] = \frac{K_{\text{instab.}} [\text{Cu(CN)}_4^{3-}]}{[\text{CN}^{-1}]^4}$$

$$= \frac{6.4 \times 10^{-15} \times 0.1}{(0.2)^4} = 4 \times 10^{-13}\text{M}$$

$$\text{Now, } [\text{S}^{2-}] = \frac{K_{\text{SP}}(\text{Cu}_2\text{S})}{[\text{Cu}^+]^2} = \frac{2.56 \times 10^{-27}}{(4 \times 10^{-13})^2}$$

$$= 1.6 \times 10^{-2}\text{M}$$

$$\therefore [\text{H}^+] = \sqrt{\frac{K_a \times [\text{H}_2\text{S}]}{[\text{S}^{2-}]}} = \sqrt{\frac{1.6 \times 10^{-21}}{1.6 \times 10^{-2}}}$$

$$= 10^{-10}\text{M and pH}=10.0$$

4. **Ans (D)**

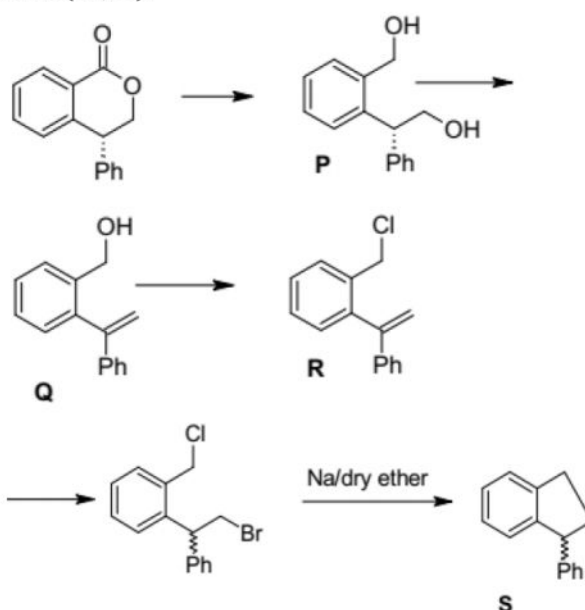
$$n = (1+4+9+\dots+n^2) = \frac{P_f(1+2+3+\dots+n)V}{RT}$$

$$P_f = \frac{P \left[\frac{n(n+1)(2n+1)}{6} \right]}{\frac{n(n+1)}{2}} = \left(\frac{2n+1}{3} \right) P$$

PART-2 : CHEMISTRY

SECTION-I (ii)

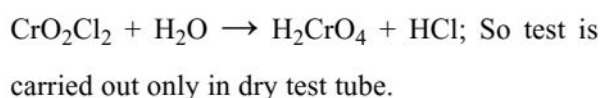
6. **Ans (A,B)**



7. **Ans (A,B,C,D)**

A) It is deep red colouration due to the formation of Fe(SCN)_3 .

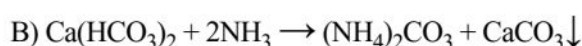
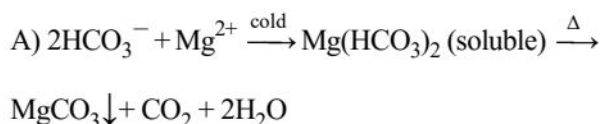
B) In presence of moisture following reaction will occur:



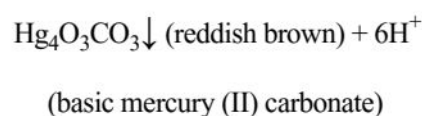
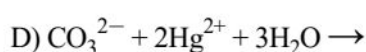
C) White precipitate of PbSO_4 is formed and hence brown ring is not visible.

D) Gives yellow precipitate of CdS .

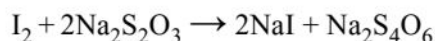
8. **Ans (B,C,D)**



C) True statement



9. Ans (A,B,D)



$$nf = 2 \quad nf = 1$$

No. of Meq in Part-I = No. of meq of hypo

$$= 15 \times 0.4$$

No. of Meq $I_2 = 6 \text{ meq} = 3 \text{ millimoles}$

Part-II



Millimoles of I_2 with NaOH

$$= \frac{(30) - (2 \times 3)}{2} = 12 \text{ millimoles}$$

Total number of millimoles of $I_2 = 12 + 3$

= 15 millimoles

$$\text{Initial molarity} = \frac{15 \text{ mmol}}{150 \text{ ml}} = 0.1M$$

PART-2 : CHEMISTRY

SECTION-III

2. Ans (4)

SO_2 , SO_3 , H_3PO_4 and $HClO_4$, has $p\pi-d\pi$ bonding

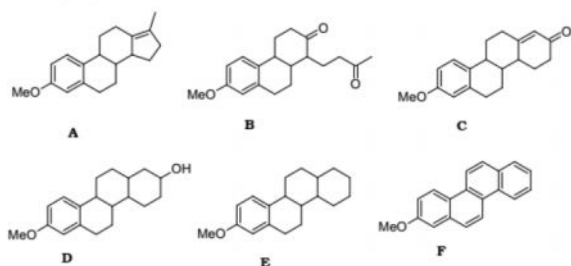
3. Ans (4)



4. Ans (8)

A) ClF_3 ; $x = 11$ B) UF_6 ; $y = 18$ C) ClF ; $z = 6$

5. Ans (7)



6. Ans (4)

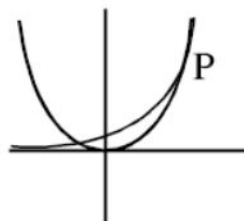
b, c, e, j only

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (D)

Since both curves touches each other



$$\Rightarrow m_1 = m_2$$

$$4(x - a) = 2e^{2x}$$

$$4(x - a) = 2.2(x - a)^2$$

$$\Rightarrow (x - a)(x - a - 1) = 0$$

$$\Rightarrow x = a \text{ or } x = a + 1$$

If $x = a$ then (I^{st}) curve $\Rightarrow y = 0$

& 2^{nd} curve $\Rightarrow y = e^{2a}$ but $e^{2a} \neq 0 \forall a \in \mathbb{R}$

$$\therefore x \neq a$$

$$\Rightarrow x = a + 1$$

$$\therefore 1^{\text{st}} \text{ curve} \Rightarrow y = 2 \therefore (x, y) = (a + 1, 2)$$

(x, y) lies on 2^{nd} curve also, $\therefore 2 = e^{2a+2}$

$$\Rightarrow 2a + 2 = \ln 2$$

$$a = \frac{1}{2}(\ln 2 - 2)$$

$$\Rightarrow -1 < a < 0$$

2. Ans (B)

Let $P(t, t^2)$ be the parametric co-ordinates of any point on the parabola $y = x^2$

$$\Rightarrow \left. \frac{dy}{dt} \right|_{(t, t^2)} = 2t$$

Hence normal at P is $(y - t^2) = \frac{-1}{2t}(x - t)$

If this line meets the curve again at $Q(t', t'^2)$,

then

$$(t'^2 - t^2) = \left(-\frac{1}{2t} \right) (t' - t)$$

$$\Rightarrow t' + t = -\frac{1}{2t}$$

$$\Rightarrow t' = -t - \frac{1}{2t}$$

If ℓ is length of the normal chord then $z = (\ell)^2$

$$= PQ^2$$

$$\Rightarrow z = (t - t')^2 [1 + (t + t')^2]$$

$$\Rightarrow z = \left(t + t + \frac{1}{2t} \right)^2 \left(1 + \left(-\frac{1}{2t} \right)^2 \right)$$

$$\Rightarrow z = \frac{(4t^2 + 1)^3}{16t^4}$$

Put $4t^2 = p$ where $p > 0$

$$z = \frac{(p+1)^3}{p^2} = (p+3) + \frac{3}{p} + \frac{1}{p^2}$$

$$\Rightarrow \frac{dz}{dp} = \left(1 - \frac{3}{p^2} \right) - \frac{2}{p^3} = 0$$

$$\Rightarrow (p+1)^2 (p-2) = 0$$

$$\Rightarrow p = 0 \text{ as } p \neq -1 \text{ and } \frac{d^2z}{dp^2} = \frac{6}{p^3} + \frac{6}{p^4} > 0$$

Hence z is minimum when $p = 2 = 4t^2$

3. Ans (D)

$$I(k) = \int_0^\infty \frac{\ell \ln x}{x^2 + kx + k^2} dx$$

Put $x = \frac{k}{t}$ we get

$$I(k) = \frac{1}{k} \int_0^\infty \frac{\ell \ln k - \ell \ln t}{t^2 + t + 1} dt$$

$$\Rightarrow I(k) = \frac{\ell \ln k}{k} \int_0^\infty \frac{dt}{t^2 + t + 1} - \frac{I(1)}{k}$$

Now $I(1) = -I(1)$

$$I(k) > \frac{\ell \ln k}{k} \int_0^\infty \frac{dt}{t^2 + t + 1} (k > 1)$$

4. Ans (D)

Because $|z| \leq 1$ and $|iw| \leq 1$ and $|z - iw| = 2$, so z, iw are end points of diameter of $x^2 + y^2 = 1$.

Similarly z and $i\bar{w}$ are also end points of same diameter (because $|z - i\bar{w}| = 2$)

$\Rightarrow iw = i\bar{w} \Rightarrow iw$ lies on real axis

$\Rightarrow z$ lies on real axis

$\Rightarrow$ either $z=1, iw=-1$ or $z=-1, iw=1$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,C)

$$\text{Let } t_r = \left(\frac{1}{r(r+1)} \right)^3 = \left(\frac{1}{r} - \frac{1}{r+1} \right)$$

$$= \frac{1}{r^3} - \frac{1}{(r+1)^3} - \frac{3}{r(r+1)} \left(\frac{1}{r(r+1)} \right)$$

$$= \frac{1}{r^3} - \frac{1}{(r+1)^3} - 3 \left(\frac{(r+1) - r}{r(r+1)} \right)^2,$$

$$t_r = \frac{1}{r^3} - \frac{1}{(r+1)^3} - 3 \left(\frac{1}{r^2} + \frac{1}{(r+1)^2} \right) + 6 \left(\frac{1}{r} - \frac{1}{r+1} \right)$$

$$\sum_{r=1}^n t_r = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{(n+1)^3} \right) - 3 \left(\frac{\pi^2}{6} + \frac{\pi^2}{6} - 1 \right) + 6 \left(1 - \frac{1}{n+1} \right)$$

$$\lim_{n \rightarrow \infty} \sum t_r = 10 - \pi^2$$

6. Ans (A,B,C)

Since $\sec^2 \theta > 1$

$$\Rightarrow [(n+1) \sec^2 \theta] > [n \sec^2 \theta]$$

Hence, f and g are both one-one

Let $k < n \sec^2 \theta < k+1$ and

$$k < m \operatorname{cosec}^2 \theta < k+1$$

$$k \cos^2 \theta < n < (k+1) \cos^2 \theta \text{ and}$$

$$k \sin^2 \theta < m < (k+1) \sin^2 \theta$$

Adding given $k < \text{integer} < k+1$

$$\Rightarrow A \cap B = \phi$$

Suppose $k \notin A$

$$\Rightarrow n \sec^2 \theta < k \text{ and } (k+1) < (n+1) \sec^2 \theta$$

$$\Rightarrow n < k \cos^2 \theta \text{ and } n+1 > (k+1) \cos^2 \theta$$

$$\Rightarrow k < (k-n) \operatorname{cosec}^2 \theta, k+1 > (k-n) \operatorname{cosec}^2 \theta$$

$$\Rightarrow [(k-n) \operatorname{cosec}^2 \theta] = k$$

$$\Rightarrow k \in B$$

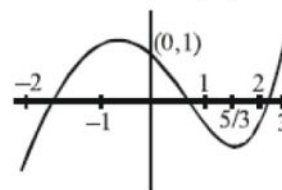
7. Ans (B,C)

Required probability =

$$\begin{aligned} & \frac{P(P_1 \text{ reaches final} \cap P_2 \text{ reaches semifinal but not in final})}{P(P_2 \text{ reaches semifinal but not in final})} \\ &= \frac{P(P_1 \text{ and } P_2 \text{ reaches semifinal} \cap P_1 \text{ wins semifinal but } P_2 \text{ loses semifinal})}{P(P_2 \text{ reaches semifinal and } P_2 \text{ loses semifinal})} \\ &= \frac{\frac{{}^{14}C_2}{{}^{16}C_4} \times \left(\frac{1}{3}\right) \left(\frac{1}{2} + \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}\right)}{\frac{{}^{15}C_3}{{}^{16}C_4} \times \frac{1}{2}} = \frac{2}{15} \end{aligned}$$

8. Ans (B,D)

$$= \left(\frac{14 \times 13 \times 12}{16 \times 15 \times 14 \times 13} \right) \left(\frac{8}{3} \right) = \left(\frac{2}{15} \right)$$



$$\text{Clearly, } \begin{vmatrix} x & x & 5 \\ 1 & x & 0 \\ 1 & 1/5 & x \end{vmatrix} = 0 \Rightarrow x^3 - x^2 - 5x + 1 = 0$$

$$\text{Assume } f(x) = x^3 - x^2 - 5x + 1$$

$$\therefore f(x) = 3x^2 - 2x - 5 = 0 \Rightarrow x = -1 \text{ \& } \frac{5}{3}$$

$$\text{Clearly } [\alpha] + [\beta] + [\gamma] = -2 + 0 + 2 = 0$$

$$\text{Now } x^3 - x^2 - 5x + 1 = (x - \alpha)(x - \beta)(x - \gamma)$$

Put $x = -1$, we get

$$(1 + \alpha)(1 + \beta)(1 + \gamma)$$

$$= -(-1 - 1 + 5 + 1) = -4$$

Now, volume of parallelepiped

$$= |x^3 - x^2 - 5x + 1| \times \in [-2, 3]$$

Maximum value will occur at $x = \frac{5}{3}$ and is equal to

$$\left| \frac{125}{27} - \frac{25}{9} - \frac{25}{3} + 1 \right| = \left| \frac{125 - 75 - 225 + 27}{27} \right| = \frac{148}{27}$$

9. Ans (B,C,D)

Orthocentre of ΔPQR lies on hyperbola

$$\Rightarrow \lambda = 3$$

Incentre of ΔABC is orthocenter of ΔPQR

$$\text{Equation of AB is } x + y - \frac{26}{5} = 0$$

10. Ans (A,C,D)

Obviously $f(1 + \alpha) = f(1 - \alpha) \forall \alpha \in (0,1)$

Domain of f is (0, 2)

Hence (A) is true

Maxima of $\ln(2x - x^2)$ as well as that of

$\sin x \left(\frac{\pi x}{2} \right)$ occur at $x = 1$.

Hence (C) is true

As $x \rightarrow 0^+$ or $x \rightarrow 2^-$, $f(x) \rightarrow -\infty$

Hence (D) is true.

PART-3 : MATHEMATICS

SECTION-III

1. **Ans (3)**

$$f(2) = 8 - (r-1)4 - 22 + 4r = -ve$$

$$f(-2) = -8 - (r-1)4 + 22 + 4r = +ve$$

⇒ One of the integral root can be, -1 or 0 or 1

Case - I : $x = -1$ is one root ⇒ $r = -\frac{11}{3}$ rejected

Case - II : $x = 0$ is one root ⇒ $r = 0$ but for $r = 0$,

$f(x) = x^3 - x^2 - 11x$ which does not have all integral roots

⇒ $r = 0$ (rejected)

Case - III : $x = 1$ is one root ⇒ $r = 3$ and at $r = 3$

all roots are integral

(Equation is $x^3 - 2x^2 - 11x + 12 = 0$)

$$\Rightarrow (x-1)(x-4)(x+3) = 0$$

2. **Ans (4)**

$$4I + 3J = \int_0^{\frac{\pi}{2}} 1 dx = \frac{\pi}{2}$$

$$3I - 4J = \int_0^{\frac{\pi}{2}} \frac{3 \cos x - 4 \sin x}{3 \sin x + 4 \cos x + 25} dx$$

$$= [\ln(3 \sin x + 4 \cos x + 25)]_0^{\frac{\pi}{2}} = \ln 28 - \ln 29$$

$$3I - 4J = \ln\left(\frac{28}{29}\right) \quad \dots(i)$$

$$4I + 3J = \frac{\pi}{2} \quad \dots(ii)$$

$$11I + 2J = \pi + \ln\left(\frac{28}{29}\right) \equiv 2\pi + b \ln\left(\frac{c}{d}\right)$$

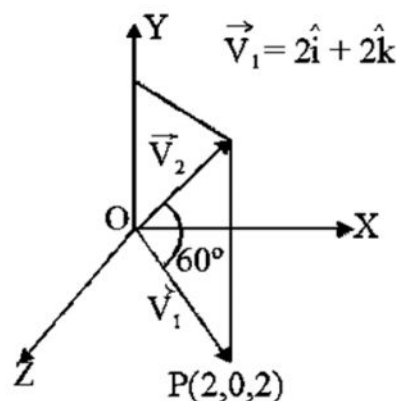
$$\Rightarrow d = 1 \text{ and } b = 1, c = 28, d = 29$$

$$\therefore a + b + c + d - 55 = 59 - 55 = 4$$

3. **Ans (4)**

Equation of plane perpendicular to XOZ plane is

$P : x - z = 0$, $\vec{V}_2$ is coplanar with $\vec{V}_1$ and $\hat{j}$.



$$\Rightarrow \vec{V}_2 = x\vec{V}_1 + y\hat{j}$$

$$\Rightarrow \vec{V}_2 = x(2\hat{i} + 2\hat{k}) + y\hat{j}$$

$$\therefore |\vec{V}_2| = |\vec{V}_1| = 2\sqrt{2}$$

$$\therefore 4x^2 + y^2 + 4x^2 = 8 \quad \dots(i)$$

$$\text{Now, } \vec{V}_2 \cdot \vec{V}_1 = |\vec{V}_2| |\vec{V}_1| \cos \frac{\pi}{3} = 4$$

$$4 = (x(2\hat{i} + 2\hat{k}) + y\hat{j}) \cdot (2\hat{i} + 2\hat{k})$$

$$4 = 8x \Rightarrow x = \frac{1}{2} \Rightarrow y = \pm\sqrt{6}$$

$$\Rightarrow \vec{V}_2 = \hat{i} + \sqrt{6}\hat{j} + \hat{k} \quad (\because \vec{V}_2 \cdot \hat{j} > 0)$$

$$\text{Now, } \vec{V}_3 = \vec{V}_1 + x\hat{i}$$

$$\vec{V}_3 \cdot \vec{V}_1 = 0 \Rightarrow 8 + 2x = 0 \Rightarrow x = -4$$

$$\Rightarrow \vec{V}_3 = -2\hat{i} + 2\hat{k}$$

⇒ Volume of tetrahedron

$$V = \frac{1}{6} \begin{vmatrix} 2 & 0 & 2 \\ 1 & \sqrt{6} & 1 \\ -2 & 0 & 2 \end{vmatrix}$$

$$\Rightarrow V = \frac{1}{6} 8\sqrt{6} \Rightarrow \frac{3V}{\sqrt{6}} = 4$$

4. Ans (3)

$$h'(x) = f(x)$$

$$h''(x) = f'(x)$$

$$h(1) = 0, f(1) = f'(1) = h'(1) = h''(1) = 1 = g(1)$$

$$\Rightarrow f(g(x)) g'(x) = 1$$

$$\Rightarrow f(g(1))g'(1) = 1 \Rightarrow f(1)g'(1) = 1 \Rightarrow g'(1) = 1$$

$$G(x) = x^2 g(x) - xh(g(x))$$

$$G'(x) = 2xg(x) + x^2 g'(x) - h(g(x)) - xh'(g(x))g'(x)$$

$$= 2xg(x) + x^2 g'(x) - h(g(x)) - x^2 g'(x)$$

$$= 2xg(x) - h(g(x))$$

$$G''(x) = 2g(x) + 2xg'(x) - h'(g(x))g'(x)$$

$$= 2g(x) + xg'(x)$$

$$\therefore G''(1) = 2g(1) + g'(1) = 2 + 1 = 3$$

5. Ans (0)

$$(x+1)^3 + (y+1)^3 = 16$$

$$y = -1 + \left(16 - (x+1)^3\right)^{\frac{1}{3}} = f(x)$$

$$x = -1 + \left(16 - (y+1)^3\right)^{\frac{1}{3}}$$

$$\therefore f^{-1}(x) = -1 + \left(16 - (x+1)^3\right)^{1/3} = g(x)$$

$$\Rightarrow f(x) = g(x)$$

$$f(-1 + \sqrt[3]{15}) = g(-1 + \sqrt[3]{15}) = 0$$

$$(f(x+g(x)).g(x+f(x))) = (f(x+f(x)))^2$$

$$\frac{d}{dx}(f(x+f(x)))^2 = 2f(x+f(x))$$

$$f'(x+f(x)).(1+f'(x))$$

$$\frac{d}{dx}(f(x+f(x)))^2 \text{ at } x = -1 + \sqrt[3]{15} = 0$$

$$\therefore g(-1 + \sqrt[3]{15}) = f(-1 + \sqrt[3]{15}) = 0$$

6. Ans (1)

$$P_1 : x + y - z = 1$$

$$P_2 : (m-1)x + (3m-7)y - 5z = -3$$

$$P_3 : (m-2)x + (2m-5)y = -4$$

$\therefore$ all the three planes intersects at a line

$$\therefore \begin{vmatrix} 1 & 1 & -1 \\ m-1 & 3m-7 & -5 \\ m-2 & 2m-5 & 0 \end{vmatrix}$$

$$\Rightarrow m = 3, -2$$

At $m = 3$

$$\left. \begin{array}{l} P_1 : x + y - z = 1 \\ P_2 : 2x + 2y - 5z = -3 \\ P_3 : x + y = -4 \end{array} \right\} \text{System does not have}$$

any solution

but at $m = -2$

$$P_1 : x + y - z = 1$$

$$P_2 : 3x + 13y + 5z = 3$$

$$P_3 : 4x + 9y = 4$$

$$\Rightarrow 5P_1 + P_2 = 2P_3$$

$\Rightarrow$ Direction ratio of line of intersection are given by

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 3 & 13 & 5 \end{vmatrix} = 18\hat{i} - 8\hat{j} + 10\hat{k}$$

$$\Rightarrow \langle 9, -4, 5 \rangle$$

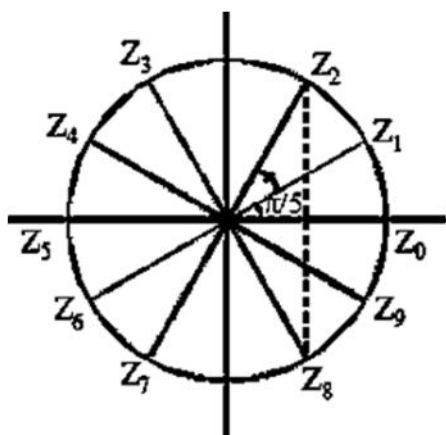
$$\Rightarrow a = 9\lambda, b = -4\lambda, c = 5\lambda$$

$$\therefore \frac{a+b}{c} = 1$$

7. Ans (8)

$$z = \cos \frac{\pi}{5} + 2i \sin \frac{\pi}{10} \cos \frac{\pi}{10}$$

$$\Rightarrow z = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5} = e^{i\frac{\pi}{5}}$$



$$\because Z_p = z^p = e^{ip\frac{\pi}{5}}, p = 0, 1, 2, 3, \dots, 9$$

$$(z_2, z_0)(z_3, z_1)(z_4, z_2)(z_5, z_3)(z_6, z_4)(z_7, z_5)(z_8, z_6)$$

$$(z_9, z_7)$$

$\Rightarrow$ 8 ordered pairs

8. Ans (7)

$$\frac{dy}{dx} = (x+1) + \frac{y-3}{x+1}$$

$$y = x^2 + x + 3 + c(x+1);$$

$$\because y_{(x=2)} = 0 \text{ so } C = -3$$

$$f(x) = x^2 - 2x$$

$$\text{Area} = \int_0^2 2x - x^2 dx = \frac{4}{3}; m+n=7$$